Gradient-Enhanced Gaussian Process Regression for Non-Intrusive POD-Based Reduced-Order Models: An Adjoint-Informed 1D Diffusion Study

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ABSTRACT

Abstract:

- Expensive coupled-physics solves → cheap surrogate. GE-GPR is the method; a 1D diffusion sandbox is used here as a verification and validation (V&V) vehicle, where closed-form/adjoint ground truth makes the method's gains checkable before deploying it on harder problems.
- New idea: an adjoint solve gives free gradients of each POD coefficient. Train a gradient-enhanced GP (GE-GPR) on those instead of a value-only GP.
- Verified on two 1D diffusion test cases (layered material properties, a smooth void/Teflon interface, maybe both combined) with exact adjoint gradients, then applied to a SCEPTRE-based secondary problem where no adjoint exists, to check the method still helps with only approximate gradients.
- Punchline: GE-GPR hits the same accuracy with [???]× fewer training solves.
- Takeaway: adjoint gradients are a cheap way to make ROM surrogates more sample-efficient. 1D diffusion is shown here due to page constraints; the same GE-GPR pipeline is targeted for the full 3D coupled-physics stack by the December review, where time allows.

Keywords: reduced-order modeling, proper orthogonal decomposition, Gaussian process regression, adjoint sensitivity, gradient-enhanced surrogate

1. INTRODUCTION

- Expensive coupled-physics solves are too slow to call over and over (design loops, UQ). Non-intrusive ROMs fix that with a cheap parameter → QoI surrogate.
- Usual recipe: POD on a QoI snapshot matrix, project to a few coefficients $\alpha(\mu)$, fit a GP to $\mu \mapsto \alpha(\mu)$.
- Problem: a plain GP only learns from function values. Every expensive solve gives one data point per mode.
- Adjoints fix that almost for free: one adjoint solve per POD mode gives the exact gradient w.r.t. every parameter at once.
- So: feed those gradients into the GP too → gradient-enhanced GPR (GE-GPR / GEK). Cite [1, 2, 3, 4, 5].
- **This paper's contribution is the method**, GE-GPR built on adjoint-derived POD-coefficient gradients. A 1D one-group diffusion solver is used as the V&V vehicle for it: closed-form/adjoint-verifiable gradients let every claim about GE-GPR's sample efficiency be checked exactly, before the same recipe is pointed at a harder, adjoint-free code (SCEPTRE, Sec. 5.1) where the gains can't be checked the same way.

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- Page-limit note: only the 1D diffusion V&V results and the SCEPTRE secondary problem fit here. The same pipeline (forward solve → POD → adjoint → GE-GPR) is not solver-specific, and is targeted for the full 3D coupled-physics stack for the December review where time allows.
- What this paper does:
  - derives the adjoint sensitivities of the POD coefficients for the diffusion solver (material layers, plus a smooth moving-interface geometry)
  - builds GE-GPR from scratch (scikit-learn has no gradient-observation support) and checks it against standard GPR
  - verifies GE-GPR against exact adjoint gradients in 1D diffusion, then applies it unchanged to a SCEPTRE-based secondary problem with only approximate (finite-difference) gradients
  - lays out a verification (identity check, FD gradient check, sweep plots) that's reusable elsewhere.

2. V&V SANDBOX: FORWARD MODEL AND BASELINE ROM

The gradient-free scaffolding GE-GPR is verified against, kept brief since it is not the contribution: 1D one-group diffusion with piecewise-constant material layers (or a smooth void/Teflon interface, Sec. 3), cell-centered 2nd-order FD → sparse system $A(\mu)\phi(\mu) = s(\mu)$, parameters μ = per-layer D, Σ_a, q. A masked flux window $y(\mu) = H\phi(\mu)$ is projected onto a rank-r POD basis Ψ (thin SVD, energy cutoff) to give coefficients $\alpha(\mu) = \Psi^\top y(\mu)$; the baseline surrogate is one ARD-RBF GP per mode (scikit-learn). Field error ε_y, compared against the POD floor $\varepsilon_{\text{proj}}$, is the metric used in every result that follows.

3. ADJOINT SENSITIVITIES OF THE POD COEFFICIENTS

The gradients that feed Section 4.

- Standard trick: a scalar QoI $J = g^\top \phi$ comes from one adjoint solve $A^\top \lambda = g$, via $J = \lambda^\top s$
- Each POD coefficient α_k is exactly this kind of scalar QoI ($g_k = H^\top \psi_k$) → adjoint solve per mode gives $\partial\alpha_k/\partial\mu_j$ for every μ_j at once
- Material-layer case: clean closed forms for $\partial\alpha_k/\partial q_m$ and $\partial\alpha_k/\partial\Sigma_{a,m}$; $\partial\alpha_k/\partial D_m$ requires more work
- Geometry case: a hard material boundary isn't differentiable at the boundary. Fix: swap it for a smooth sigmoid switch at the one moving interface (void/Teflon), derive the chain rule through it (two knobs: interface location, transition sharpness)
- Same adjoint machinery works for both cases; geometry just swaps a 0/1 layer indicator for a smooth per-cell direction vector
- Verification, same three checks each time: identity check ($\lambda^\top s$ vs. direct projection, matches to solver tolerance), finite-difference gradient check, and a 1D sweep with a tangent-line overlay

4. GRADIENT-ENHANCED GAUSSIAN PROCESS REGRESSION

The method contribution of this paper.

- Core idea: a GP is closed under differentiation, so the joint distribution of $\alpha_k(\mu)$ and $\partial\alpha_k/\partial\mu$ at the training points is itself Gaussian, with covariance blocks given in closed form by differentiating the kernel. Training on both is therefore not an approximation or a regularizer bolted onto a value-only GP – it is the same GP model, conditioned on more of what it already implies.
- Why this construction and not, e.g., feeding gradients in as extra scalar features or fitting a separate model per derivative channel: those break the single joint-covariance structure (channels would be modeled as if independent, throwing away the $\text{Cov}(\alpha_k, \partial\alpha_k/\partial\mu_m)$ correlation that is exactly what makes a gradient observation informative about nearby function values). Stacking y_{aug} =

$[\alpha_k; \partial\alpha_k/\partial\mu]$ into one vector against one augmented kernel is what keeps that correlation in the model.

- Concretely: value-value, value-gradient, gradient-gradient covariances (closed form for ARD-RBF, Appendix A). Bigger matrix ($M(1 + p_{\text{act}})$ instead of M), fit by maximizing the augmented marginal likelihood – the same hyperparameters ($\sigma_f, \ell_1, \ldots, \ell_{p_{\text{act}}}$) now have to explain both the values and their slopes simultaneously, which is what lets a handful of gradient observations pin down length scales that would otherwise need many more function evaluations.
- Cost/conditioning tradeoff, stated plainly: the augmented Gram matrix grows cubically in p_{act} (factorization cost $O((M(1 + p_{\text{act}}))^3)$) and can become ill-conditioned when length scales shrink relative to the training spacing, since the gradient-gradient block's magnitude scales as $1/\ell^2$. This bounds how far the method scales in p_{act} as-is (Sec. 6).
- Built from scratch, since scikit-learn's GP can't take derivative observations.
- Everything else about the surrogate matches the baseline (one GP per mode, same kernel family). Isolates the effect of adding gradients.
- Same code reused unmodified on the geometry case, just with a different sampler/adjoint front end feeding it, and again unmodified on the SCEPTRE secondary problem (Sec. 5.1) with finite-difference gradients standing in for an adjoint. The method doesn't care what μ physically means or where the gradients came from – only that they are (approximately) $\partial\alpha_k/\partial\mu$.

5. RESULTS

- **Material-layer case (V&V):** SVD spectrum/rank choice, projection-error floor vs. r, then a learning curve comparing GE-GPR vs. standard GPR at matched training-set sizes. Report as "GE-GPR at M=· matches standard GPR at M=·" [???].
- **Geometry case (V&V):** Run once with only the interface location active (close to a 1D problem with nearly all the SVD energy in mode 1), and once with both parameters active. Compare GE-GPR's benefit in each.
- **Combined case:** All 5 parameters active simultaneously – void_thickness, steepness, and the Teflon shell's D, Σ_a, q – in one shared forward/adjoint/GE-GPR pipeline (one adjoint solve per mode still gives all 5 partials at once) instead of the 5 one-at-a-time studies above. At matched M=80: GE-GPR median $\varepsilon_y = 5.8 \times 10^{-4}$ vs. standard GPR's 2.9×10^{-3} (80% reduction, vs. 0–95% across the 5 individual one-at-a-time studies at the same bounds) – gradient information compounds faster than function values as p_{act} grows, since one adjoint solve still yields all 5 partials while a value-only GP needs proportionally more samples to resolve the higher-dimensional map. GE-GPR also lands closer to the POD floor ($\varepsilon_{\text{proj}} = 2.0 \times 10^{-4}$, so GE-GPR is $\sim 3\times$ the floor vs. standard GPR's $\sim 15\times$).
- **Cost accounting:** An adjoint solve reuses the same factorized matrix as the forward solve, so it's cheap per sample; the trade-off is a bigger, pricier-to-fit kernel matrix.

5.1. SCEPTRE Secondary Problem

Unlike the diffusion cases above, SCEPTRE is a real, export-controlled coupled-physics code with no adjoint/sensitivity capability – the point of including it is to check whether GE-GPR still helps when the "gradient" fed to it is only an approximation, not the exact adjoint value used for V&V above.

- Problem: net SGEMP direct-drive charge $q_{\text{net}}(\mu)$ as a function of void-region thickness, a scalar QoI from SCEPTRE forward flux dumps (no POD stage needed for a scalar output).
- No adjoint available $\rightarrow$ the gradient $\partial q_{\text{net}}/\partial(\text{void thickness})$ used by GE-GPR is a finite-difference approximation across the sorted training cases – explicitly a rough, honestly-labeled placeholder, not the exact-gradient regime the diffusion V&V cases establish.

- Same `GEGaussianProcess` code, unmodified, as the diffusion cases – only the front end supplying $(X, y, \text{gradient})$ changes.
- Result [???]: GE-GPR vs. standard GPR on the same SCEPTRE training cases, [N] cases total.
- Takeaway this is meant to support: adjoint gradients are the clean case, but GE-GPR degrades gracefully to FD-approximated gradients on a real code that was never built with an adjoint in mind – a practical-applicability argument the 1D V&V cases alone can't make.

6. DISCUSSION

- Gradient enhancement helps most when: parameter count is small, the forward solve is the expensive part.
- Helps least when: the coefficient map is already easy to learn from values alone (the simple geometry case)
- Known limitation: the augmented kernel grows with $M(1 + p_{\text{act}})$. Doesn't scale to large p as-is.
- Combined 5-parameter case: gradient enhancement's relative benefit *grew* with p_{act} (80% error reduction at p_{act}=5 vs. 0–95% across the individual one-at-a-time studies) – consistent with the "helps most when parameter count is small" framing above only insofar as it's about the forward-solve/adjoint-solve cost ratio, not the raw dimension count: each adjoint solve still amortizes over all 5 partials, so more active parameters means more free information per expensive solve, not less.
- SCEPTRE case: even an approximate (FD) gradient appears to help [???] – read together with the diffusion V&V results, this suggests the exact-adjoint requirement is not a hard prerequisite for GE-GPR to be worth trying on a new code.

7. CONCLUSIONS AND FUTURE WORK

- Restate the contributions from Sec. 1, past tense, with real numbers filled in: GE-GPR verified against exact adjoint gradients on two 1D diffusion V&V cases, then shown to carry over to a real, adjoint-free code (SCEPTRE).
- Future work: Pushing past $p_{\text{act}} \in \{1, 2\}$, maybe using the adjoint gradients themselves as a sensitivity/dimension-reduction screen; a multi-output version instead of one GP per mode; more SCEPTRE cases as they become available.
- Full 3D coupled-physics stack: this 1D study is the V&V step for that larger effort; extending the same pipeline to 3D is targeted for the December review, time permitting.

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I have no idea, thank you Dr. Ragusa and Dr. Morel?

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APPENDIX A. AUGMENTED KERNEL DERIVATIVE IDENTITIES

Full algebra behind the closed-form K_{fd}, K_{dd} terms in Sec. 4, for anyone re-implementing `GEGaussianProcess` from scratch.

APPENDIX B. VERIFICATION FIGURES

Overflow plot dump if space runs out in the main body: identity checks, FD gradient checks, for both test cases.